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September 2026

Current Affairs

8.1 Five new entries in India's UNESCO Tentative List

PRELIMS + MAINS

What: Five proposals added, total 78: four in Madhya Pradesh and one serial transnational proposal with South Africa.

Bagh Cave Paintings → nine rock-cut Buddhist caves on the Baghni River, Vindhyas, 4th-6th century CE, continuity with Ajanta.

Udaigiri Caves → 20 rock-cut caves, late 4th-early 5th century, Gupta; Varaha relief (Cave 5), Anantasayana Vishnu (Cave 13); Shaiva and Jain elements.

Ginnorgarh Fort → Raj Gond hill fort and palace complex, early 16th-early 18th century; rainwater harvesting.

Significance → inclusive heritage, cultural diplomacy, conservation planning, Article 49 and Article 51A(f).

POINTS

→ Tentative List = inventory of properties with potential Outstanding Universal Value; mandatory first step to World Heritage listing.

→ Group of Temples of the Chambal Valley: Bateshwar (200+ small shrines), Padavali, Kakarmath; 8th-11th century; Gurjara-Pratihara and Kachchhapaghata; Nagara evolution.

→ Sites of Satyagraha (India-South Africa, nine sites): Pietermaritzburg station (1893), Phoenix Settlement (1904, Indian Opinion), Champaran Satyagraha Smarak (1917; 48-foot Chunar-stone pillar), Sabarmati Ashram (Dandi March 1930), Mani Bhavan (Mumbai base 1917-34), Harijan Sevak Sangh Delhi (1932, after Poona Pact), Sevagram (1936), Aga Khan Palace Pune (1942-44), Gandhi Smriti.

PRELIMS FODDER

- World Heritage Convention adopted 1972, in force 1975, India ratified 1977; India has 45 World Heritage Sites; Committee has 21 States Parties; at least one of 10 criteria.

MAINS FODDER

- Keywords: OUV, serial transnational heritage, Nagara architecture, rock-cut architecture.

PYQ: 2018 Mains: Safeguarding Indian art heritage is the need of the moment.

8.2.1 Rajghat excavations (Varanasi)

PRELIMS

What: BHU opened 22 boxes with over 10,000 artefacts from excavations begun 1957 and started genetic and bio-archaeological study of a skeleton. This is re-examination of old material, not a new excavation.

Finds → brick structures, streets, wells, bathing tanks; pottery, terracotta, coins, copper, ivory, bone; seals/sealings naming Varanasi and officials; some depict Greek deities; occupation from first millennium BCE to medieval.

POINTS → Near the Ganga-Varuna confluence; noticed in 1939 (Kashi station expansion); trial dig 1940 by ASI's Krishna Deva; BHU excavations 1957-69 (A.K. Narain, T.N. Roy). Earliest settlement about 800 BCE, pre-NBPW layers.

8.2.2 Thirukkural

PRELIMS

What: PM gifted a Russian translation to the Russian President ahead of BRICS 2026.

POINTS → By Thiruvalluvar; 1,330 couplets (kurals of seven words) in 133 chapters of 10; three parts: Aram (virtue), Porul (polity/wealth), Inbam (love); composed between 300 BCE and 5th century CE; among the Patinenkilkanakku (Eighteen Minor Works); translated into 60+ languages.

8.2.3 Poompuhar underwater excavations

PRELIMS

What: 18-day exploration by Tamil Nadu State Department of Archaeology with Indian Maritime University found a West Asian torpedo jar fragment, four terracotta ring wells and over 250 ceramic pieces.

Torpedo jar

→ Persian Gulf/Indian Ocean trade (wine, olive oil); glazed pottery: Iran/West Asia; paddle-impressed pottery: Gangetic links; ring wells: submerged habitation.

POINTS

→ Poompuhar (Puhar, Kaveripoompattinam), Mayiladuthurai district, Cauvery mouth; early Chola port and possibly capital; carbon-dated log shows settlement by 5th-4th century BCE; areas Maruvurppakkam, Pattinappakkam, Nalangadi; mentioned in Pattinappalai, Silappadikaram, Manimekalai.

8.2.4 Ramanathaswamy Temple

PRELIMS

What: Cultural backdrop at the 18th BRICS Summit in New Delhi.

POINTS

→ Rameswaram, Ramanathapuram district; Shiva; one of 12 Jyotirlingas; Char Dham site and the only Shiva temple among the four (Badrinath, Dwarka, Puri); Dravidian architecture; 1,212 pillars in the outer corridor; 22 theerthams; praised by Appar and Tirugnana Sambandar.

8.2.5 Gollala Gudi Temple: Monument of National Importance

PRELIMS

What: ASI declared the Kakatiya-period (12th-13th century) temple at Palampet, Mulugu district, Telangana, near Ramappa Temple, a Monument of National Importance.

Trikutalaya

→ three sanctums around a central mandapa.

AMASR Act 1958

→ monument at least 100 years old with special importance; ASI under Ministry of Culture; prohibited area 100 m and regulated area a further 200 m; Article 49 directs protection.

8.3 Personalities in news

PRELIMS

What: Anniversaries, September 2026.

POINTS

→ Maharani Avantibai Lodhi (1831-1858), 195th anniversary: born Mankehdi, Seoni (MP); queen of Ramgarh; in 1857 mobilised forces in the Mandla-Ramgarh region and defeated Waddington at the Battle of Kheri; in March 1858 chose death over surrender.

→ Saratchandra Chattopadhyay (1876-1938), 150th anniversary: Bengali novelist; Devdas, Parineeta, Srikanta, Charitraheen, Pather Dabi, Sesh Prashna; Pather Dabi banned by the colonial government in 1927; Jagattarini Gold Medal 1923.

→ Dr S. Radhakrishnan (1888-1975), 138th anniversary: born Tiruttani; first Vice-President (1952-62), second President (1962-67); Ambassador to the USSR; VC of Andhra University and BHU; Oxford; works include Indian Philosophy, The Hindu View of Life; Bharat Ratna 1954; birthday 5 September is Teachers' Day.

→ Acharya Vinoba Bhave (1895-1982), 131st anniversary: Sarvodaya thinker; born Gagode, Raigad; first Individual Satyagrahi (1940); Bhoodan Movement at Pochampally, Telangana (1951), later Gramdan; "Jai Jagat"; Gita Pravachane, Swarajya Shastra; first Ramon Magsaysay Award for Community Leadership (1958); posthumous Bharat Ratna 1983.

→ Dr Bhupen Hazarika (1926-2011), 100th anniversary: born Sadiya, Assam; singer, lyricist, composer, filmmaker; "Sudhakantha", Bard of the Brahmaputra; Bistirno Parore, Manuhe Manuhor Babe; independent Assam MLA 1967-72; Dadasaheb Phalke Award 1992; posthumous Bharat Ratna 2019.

9.1 Ethics of slowing down AI

MAINS

What: Anthropic's CEO called for AI firms to "pace the frontier" until safety checks catch up.

Stakeholder dilemmas → companies (competition vs responsibility), governments (innovation vs public safety), citizens (benefits vs involuntary risks), developing countries (safety vs equal opportunity), future generations.

Ethical concerns → precaution vs innovation (precautionary principle, non-maleficence; Asilomar 1975 after the 1974 pause); duty to benefit (beneficence; AlphaFold, 2024 Chemistry Nobel); collective action dilemma (FLI 2023 six-month pause letter); self-regulation vs public accountability (independent evaluators with access); distributive and intergenerational justice (India's AI Governance Guidelines 2025 "Innovation over Restraint"; New Delhi Declaration on AI Impact 2026).

Way forward → risk-based safeguards, purposeful pauses, independent scrutiny, coordinated governance, inclusive decisions.

MAINS FODDER

- Thinkers: Hans Jonas (ethics of responsibility); Collingridge dilemma; utilitarianism and duty ethics.
- Keywords: precautionary principle, non-maleficence, beneficence, collective action problem, probity, responsible innovation.

9.2 Ethics of AI in health research

MAINS

What: September 2026 WHO report urged stronger ethical oversight; it cites ICMR Ethical Guidelines for AI in Biomedical Research and Healthcare (2023).

Dilemmas → informed consent vs secondary data use (Royal Free-DeepMind 1.6 million records, 2015); accuracy vs fairness (US algorithm using past spending underestimated Black patients' needs); beneficence vs non-maleficence (sepsis tool missed about two-thirds of cases, 2021 study); problem of many hands; health data colonialism and ethics dumping (Kenyan labellers paid under $2/hour, 2023).

Way forward → meaningful consent, bias and safety audits, skilled ethics committees, clear accountability, fair benefit-sharing.

MAINS FODDER

- Kant (ends, not means); Rawls (benefits to least advantaged); Belmont Report 1979 (respect for persons, beneficence, justice); Gandhi's "Science without humanity". Case: Havasupai tribe (US, 2010).
- Keywords: Informed consent; Ethics dumping

9.3 Dark patterns and ethics of consumer choice

MAINS

What: CCPA imposed Rs 10 lakh penalty on Rapido (September 2026) over misleading payment prompts. ASCI-Parallel study 2024 found dark patterns in 52 of 53 popular apps. CCPA Guidelines 2023 specify 13 types.

Way forward → transparent offers, equal ease of choice, ethics by design, internal accountability, effective enforcement.

POINTS → Persuasion vs manipulation (false urgency timers, confirm shaming); formal consent vs meaningful choice (subscription trap); transparency vs asymmetry (drip pricing); profit vs vulnerability (health-tech apps averaged 8.8 instances); commercial targets vs professional integrity (basket sneaking).

MAINS FODDER

• Kant; Gandhi's "Commerce without morality"; Thaler and Sunstein, Nudge. Cases: Amazon Prime FTC $2.5 billion (2025); Google and Facebook France cookie fines (2022); Zepto Rs 7 lakh CCPA penalty (2025).
• Keywords: Choice architecture

10.1 72nd National Film Awards; Dadasaheb Phalke Award

PRELIMS

What: Presented at Ekta Nagar, Gujarat, for films certified in 2024.

Best Feature Film → Article 370; Direction: Rajkumar Periasamy (Amaran); Actor: Mammootty (Bramayugam) and Kartik Aaryan (Chandu Champion); Actress: Yami Gautam (Article 370); Dadasaheb Phalke Award: Anant Nag.

POINTS → Awards instituted 1954 as State Awards for Films. Phalke Award instituted 1969; Dadasaheb Phalke directed Raja Harishchandra (1913); first recipient Devika Rani.

10.2 Ramon Magsaysay Award 2026

PRELIMS

What: Recipients: Tommy Koh (Singapore), Bo Kyi (Myanmar), Runa Khan (Bangladesh).

POINTS → Established 1957 in memory of the seventh Philippine President; by Rockefeller Brothers Fund trustees; Foundation in Manila. Vinoba Bhave first Indian (1958); Educate Girls first Indian organisation (2025).

10.3 Tenzing Norgay National Adventure Awards 2024

PRELIMS

What: India's highest adventure honour; categories Land, Water, Air and Lifetime Achievement; conferred by the President.

10.4 Hindi Diwas (14 September)

PRELIMS

What: Marks adoption of Hindi as official language after the Munshi-Ayyangar Formula.

POINTS → Hindi in Devanagari is Official Language of the Union; the Constitution names no national language. English allowed for 15 years with Parliament empowered to extend; Official Languages Act 1963 continued English beyond 1965; 1967 amendment allowed it indefinitely.

10.5 Angh (Naga film) at TIFF 2026

PRELIMS

What: Won the Platform Prize; eligible for consideration for the 2027 Academy Awards International Feature category; about a Konyak village chief in Nagaland.

2.1 Flexible Inflation Targeting (FIT): 10 years

PRELIMS + MAINS

What: India's Flexible Inflation Targeting framework completed 10 years.

FIT → RBI uses interest rates to keep inflation within a publicly announced range while supporting growth. Legal basis: RBI Act, 1934 amended in 2016.

Results → average CPI inflation fell from 8.1% (2006-16) to 4.6% (2016-26); average GDP growth 6.8% to 7%. Inflation stayed within the band three-fourths of the time in 2016-21 and two-thirds in 2021-26.

Concerns → weak transmission to lending rates; flat Phillips curve (2012-2026) so rate hikes may cost jobs with little fall in inflation, raising stagflation risk; about 90% of workforce informal; inflation mostly supply-driven (crop failure, heatwaves, crude).

Way forward → Government absorbs supply shocks, RBI prevents second-round effects; extend benchmark-linked rates to NBFCs and microfinance; bring inflation closer to 4%; cold chains, warehousing, APMC reform, rule-based buffer stock release.

POINTS → Centre, in consultation with the RBI, sets the target every five years, based on headline CPI. Retained at 4% with a +/-2% band for April 2026 to March 2031. Inflation outside the band for three consecutive quarters is a failure. The MPC sets the repo rate.

→ RBI Discussion Paper (August 2025): headline vs core (food and fuel are over 37% of the consumption basket; only Uganda targets core); 4% still right (advanced economies about 2%, major developing 3-6%); band should be guided by threshold inflation (upper) and the rate below which low inflation discourages production (lower).

PRELIMS FODDER

- MPC: statutory body under the RBI Act, 2016; 6 members (3 RBI, 3 Government nominees); majority vote; Governor has the casting vote.
- Headline inflation = total; core inflation excludes food and energy.
- Phillips curve: inverse relation between inflation and unemployment. Stagflation: high inflation, stagnation and unemployment together.

MAINS FODDER

- Urjit Patel Committee (2014): CPI as nominal anchor, 4% target with +/-2% band, committee-based decisions.
- Keywords: Price stability; Credible nominal anchor; Inflation expectations; Flat Phillips curve; Supply-side inflation

PYQ: 2017 Prelims: MPC decides the benchmark rates; it is not a 12-member body and not chaired by the Finance Minister.

2.2 Surge in inflows under RBI's forex swap facility

PRELIMS + MAINS

What: Inflows under the RBI special USD-INR swap facility reached $143.596 billion (data to 18 September 2026).

Swap is two-legged → spot leg (banks give dollars, RBI gives rupees) and reverse leg at maturity.

Other RBI steps → dollar sales and sell-buy swaps; special dollar window for IOCL, BPCL, HPCL; restrictions on forex derivatives (exposure threshold cut from $100 million to $5 million; cancelled contracts cannot be rebooked); Forex Risk Reserve (20% requirement on specified derivatives above $2 million); non-deliverable rupee contracts only for genuine hedging.

Why inflows matter → rupee stability, reserves, financing the CAD, import security (crude, fertiliser), lower imported inflation (but excessive appreciation hurts exports).

Concerns → oil import dependence; volatile capital flows (US yields); rupee weakened beyond Rs 90 per dollar in early 2026; nearly $133 billion of the $143.596 billion came from FCNR(B), a temporary buffer with repayment obligations; surplus liquidity of Rs 11.16 lakh crore in September 2026.

Way forward → diversify energy imports; stable long-term FDI and deeper bond markets; exchange-rate flexibility with targeted intervention; reserve adequacy checks; sterilisation through VRRR, SDF and OMO.

POINTS → RBI introduced special USD-INR swap facilities in June 2026 for fresh FCNR(B) deposits and specified ECBs and OFCBs. FCNR(B) window: fresh deposits with tenor 3 to 5 years; closed in August 2026.

PRELIMS FODDER

- Reserves: Foreign Currency Assets, Gold, SDRs, Reserve Tranche Position in IMF.
- FCNR(B) and ECBs operate under FEMA, 1999.

MAINS FODDER

- Impossible Trinity: free capital mobility, managed exchange rate and monetary autonomy cannot all be held.
- 2013 taper tantrum: FCNR(B) and overseas-borrowing swap windows mobilised about $34.3 billion.
- Keywords: Sterilised intervention; Reserve adequacy; Monetary trilemma; Debt-creating inflows; Rollover risk

2.3 Global bond sell-off and rising sovereign yields

PRELIMS + MAINS

What: Long-term yields in the US, UK, France and Japan hit multi-decade highs on inflation worries and fiscal pressure.

Why it matters → costlier government borrowing, less fiscal space, higher corporate and household rates, pressure on equities, risk to banks and pension funds.

For India → FPI outflow risk, rupee pressure, imported inflation, higher G-sec yields, tougher balance for RBI.

POINTS

→ US 10-year Treasury yield touched 5.34%, highest since 2002, with the sharpest quarterly rise since 1994 in Q3 2026. UK 30-year gilt crossed 6%, highest since 1998. France and Japan yields also rose.

→ Bond prices and yields move in opposite directions. Causes: persistent inflation, high interest-rate expectations, fiscal pressure (more bond supply), higher energy prices.

PRELIMS FODDER

- 1 basis point = 0.01 percentage point.
- Real yield = nominal yield adjusted for expected inflation. Yield curve = yields across maturities. Quantitative easing = central bank asset purchases. Fisher effect: nominal yield = real yield + expected inflation.

MAINS FODDER

- Structural shift away from the post-2008 era of very low rates and QE; global spillover to emerging economies even without domestic rate hikes.
- Keywords: Real yield; Yield curve; Imported inflation; Quantitative easing; Fiscal space

PYQ: 2021 Prelims: Indian G-sec yields are influenced by US Fed actions, RBI actions, and inflation and short-term rates (all three).

2.4 Annual Survey of Industries (ASI) 2024-25

PRELIMS + MAINS

What: MoSPI released ASI results for FY 2024-25.

Top factories → Tamil Nadu, Gujarat, Maharashtra. Tamil Nadu, Maharashtra, Gujarat, Uttar Pradesh and Haryana account for about 56% of manufacturing employment. Top GVA states: Maharashtra, Gujarat, Tamil Nadu, Karnataka, Uttar Pradesh. Top GVA industries: basic metals, motor vehicles, chemicals, pharmaceuticals, food products. Top employers: food products, textiles, basic metals, motor vehicles.

Positives → employment growth 7.19% (5.92% last year); GVA 9.59% with input costs (7.47%) slower than output (7.81%); wages 12.08% faster than GVA and profits (7.73%); fixed capital 10.54% (capital deepening).

Concerns → regional and sectoral concentration (five industries over 45% of GVA); capital growing faster than output and jobs (capital intensity may limit jobs).

Way forward → reward domestic value addition in PLI; industrial corridors and plug-and-play parks in lagging states; support labour-intensive sectors (food processing, textiles, footwear, toys); skilling and MSME technology.

POINTS → Conducted by NSO under MoSPI; collection under the Collection of Statistics Act, 2008 (amended 2017). Covers factories under the Factories Act, 1948, bidi and cigar units, non-CEA electricity undertakings, and units with 100+ employees on the Business Register. Factory threshold used: 10+ workers with power or 20+ without. OSH Code 2020 raises the licence threshold to 20 with power and 40 without.

PRELIMS FODDER

- ASI = annual structure of registered manufacturing; IIP = short-term volume changes; ASUSE = unincorporated non-agricultural enterprises.

MAINS FODDER

- Examples: PLI and mobile phones; Tamil Nadu dispersed industrialisation and women in factory jobs. China and Vietnam manufacturing about 25% of GDP.
- Keywords: Capital deepening; Labour-intensive manufacturing; Regional industrial imbalance; Productivity-wage nexus

2.5 District Mineral Foundations (DMF) and PMKKKY

PRELIMS + MAINS

What: The CAG flagged irregularities in DMF functioning and PMKKKY implementation in Odisha.

DMF → non-profit trust set up by States in every mining-affected district. Legal basis: Section 9B of the MMDR Act, 1957 (inserted by the 2015 Amendment). Funding: 30% of royalty (leases before 12 January 2015) or 10% (auction leases after that date). Two tiers: Governing Council and Managing Committee. Guided by the Fifth and Sixth Schedules, PESA 1996 and FRA 2006.

Status → DMFs in 656 districts of 23 States; over Rs 1.30 lakh crore collected; top States Odisha, Jharkhand, Chhattisgarh.

CAG concerns → diversion to ineligible items (wage payments under MGNREGS, government vehicles); no perspective or annual plans; endowment funds not created; about 54% fund utilisation; no social audits.

Way forward → rule-based allocation; perspective and annual plans; link DMF portals with WAMIS; collect dues with penal interest; mandatory social audits.

POINTS → PMKKKY (2015, Ministry of Mines): implemented by DMFs using DMF funds. At least 70% to high-priority sectors (drinking water, pollution control); up to 30% to other priority sectors (infrastructure, irrigation); at least 70% of funds in directly affected areas.

PRELIMS FODDER

- MMDR Act, 1957: Union regulates major minerals; States regulate minor minerals (sand, gravel) by rules. National Mineral Exploration and Development Trust set up under the 2015 Amendment.

MAINS FODDER

- Principle: those who bear the costs of mining should share its benefits. Cases: Project SAMARTHYA (Gujarat, youth training), Project Aangan-Chatra (Jharkhand, 40 anganwadis).
- Keywords: District Mineral Foundation; Royalty; Social audit; Endowment fund

2.6 Western Dedicated Freight Corridor (WDFC) completed

PRELIMS + MAINS

What: The PM inaugurated the final sections of the WDFC, completing the 2,843-km DFC network (Eastern and Western).

Western DFC 1,506 km	→ Dadri (UP) to JNPT (Maharashtra), JICA yen loans. Eastern DFC 1,337 km: Ludhiana (Punjab) to Sonnagar (Bihar), partly World Bank financed (initially planned to Dankuni, West Bengal). Implementing agency: DFCCIL (SPV under the Ministry of Railways, established 2006 under the Companies Act).
Significance	→ rail freight about Rs 1.90 per tonne-km against up to Rs 3.78 by road; decongests passenger lines; supports DMIC and Amritsar-Kolkata corridor; emissions 22 g vs 62 g CO2 per tonne-km (rail vs road); supports PM GatiShakti.
Gaps	→ terminal congestion and stabling; PPP failure (Sonnagar-Dankuni); average DFC speed 37 km/h in 2024-25 (38 in 2023-24); holding line gaps.
Way forward	→ fast-track future corridors; last-mile multimodal links (MMLPs, ICDs); private freight terminals; IoT tracking and predictive maintenance.
POINTS	→ India became the first country to run double-stack container trains under overhead electric traction on a dedicated freight corridor. → Union Budget 2026-27 proposed an East-West DFC from Dankuni (West Bengal) to Surat (Gujarat); other proposed: East Coast, North-South.

PRELIMS FODDER

• National Rail Plan Vision 2030: raise rail freight modal share to 45%. Modal share now: rail 18%, road 71% (NITI Aayog, 2021). Services: Trucks on Train, FOIS.

MAINS FODDER

• Keywords: Freight decongestion; Logistics cost; PM GatiShakti; National Logistics Policy

PYQ: 2021 Mains: Investment in infrastructure is essential for rapid and inclusive growth.

2.7 Consumer Protection (E-Commerce) Amendment Rules, 2026

PRELIMS + MAINS

What: The Department of Consumer Affairs notified amendments to the 2020 E-Commerce Rules; in force from January 2027.

Complaints	→ acknowledged in 48 hours and resolved in a month (continues); consumer gets a copy of the recorded complaint; every e-commerce entity must become a National Consumer Helpline (NCH) convergence partner.
Search and listings	→ no manipulation of search results; main ranking factors explained in order of importance; sponsored listings clearly marked.
Dark patterns	→ follow the 2023 Guidelines; yearly self-audit and a prominently displayed compliance certificate.
Disclosures	→ country of origin, best-before date, return-shipping cost, delivery and payment terms; seller GSTIN or MSME registration. Express consent needed for promotional use of consumer information.
Need	→ online shoppers about 290-300 million; about 29% of NCH complaints in 2025 were e-commerce (5.11 lakh of 17.72 lakh).
Issues	→ self-audit may not be enough; enforcement at scale; overlap with competition, data protection (DPDP Act 2023, Rules 2025) and legal metrology; burden on small businesses.
Way forward	→ CCPA audit standards and continuous compliance; automated and risk-based supervision; coordination among CCPA, CCI, MeitY and Legal Metrology.
POINTS	→ Price cuts must show the prior price, meaning the lowest price in the preceding 30 days.

PRELIMS FODDER
- Marketplace model: platform facilitates transactions; inventory model: entity owns and sells the stock.
- CCPA: Section 10, Consumer Protection Act, 2019; set up 2020, New Delhi.
- CCI Market Study (2020) flagged platform neutrality, price parity, deep discounts.
- EU also uses a 30-day lowest-price benchmark.

MAINS FODDER
- Keywords: Dark patterns; Choice architecture; Information asymmetry; Platform accountability

PYQ: 2022 Prelims: foreign-owned e-commerce firms may not sell own goods as inventory and are limited in owning large sellers.

2.8 DILRMP 3.0 (2026-31)

PRELIMS + MAINS

What: The Ministry of Rural Development issued Operational Guidelines for DILRMP 3.0 (2026-2031).

History → NLRMP (2008-16) from merging CLR and SRA&ULR; DILRMP (2016-26) as a Central Sector Scheme with Bhu-Aadhaar (ULPIN) and NGDRS. DILRMP 3.0: Central Sector Scheme, outlay Rs 565.50 crore over FY 2026-27 to 2030-31, run by the Department of Land Resources; shift from digitisation to an integrated, interoperable GIS-based Digital Public Infrastructure.

Achievements → 99.90% Records of Rights digitised, 97% cadastral maps, 99% Sub-Registrar Offices computerised.

Pillars → Land Stack (federated, links cadastral maps, RoR, registration and court cases through APIs); Bhu-Aadhaar (14-digit geospatial ID for every parcel); Registration Seva Kendras (75 high-footfall SROs, like Passport Seva Kendras); NAKSHA (urban survey, urban property cards).

Need → over 60% of civil litigation is land related and takes about 20 years; fragmented databases; unclear titles delay projects; small farmers lack verifiable titles for loans.

Challenges → digitally signed RoRs not legally valid in all States; presumptive titling, not guaranteed title; legacy errors; NAKSHA still at pilot stage.

Way forward → legal validity for digital RoRs; conclusive titling (Rajan committee on Financial Sector Reforms; NITI Aayog draft Model Land Titling Act with Land Titling Authority and Appellate Tribunals); verify legacy records; auto-triggered mutation after registration.

PRELIMS FODDER
- SVAMITVA (2020, Ministry of Panchayati Raj): property cards for village abadi areas. Cadastral map: large-scale map of land parcel boundaries and ownership. Mutation: updating land records after sale, inheritance, partition or court order.

MAINS FODDER
- Keywords: Bhu-Aadhaar (ULPIN); Land Stack; Digital Public Infrastructure; Conclusive titling

2.9.1 MDR on high-value UPI merchant payments

PRELIMS

What: NPCI announced a Merchant Discount Rate on select high-value person-to-merchant (P2M) UPI payments under the Payment and Settlement Systems Act, 2007. UPI stays free for P2P.

MDR → fee a merchant pays its bank for accepting digital payments, as a percentage of value; shared among acquiring bank, issuing bank, UPI app and NPCI.

Framework → P2P free (any amount); P2M up to Rs 2,000 free; above Rs 2,000 0.4%; capped at Rs 300 for Rs 75,000 and above; essential sectors (railways, telecom, insurance, fuel, utilities) above Rs 2,000 flat Rs 5.

POINTS

→ A Small Merchant Fund from part of MDR revenue for Tier 3-6 centres, North-East, J&K, Ladakh, and PM SVANidhi and PM Vishwakarma beneficiaries. Merchants cannot pass MDR to customers.

MAINS FODDER

• Keywords: MDR; P2M; PSS Act 2007; UPI sustainability

2.9.2 National Investment and Infrastructure Fund (NIIF)

PRELIMS

What: NIIF Infrastructure Fund II has raised about Rs 19,000 crore towards a Rs 30,000 crore target. Fund I had about Rs 16,000 crore.

Funds

→ Master fund, fund of funds (e.g. India-UK Green Growth Equity Fund), Strategic Opportunities Fund, India-Japan Fund.

POINTS

→ NIIF is a sovereign-anchored fund (government a key investor, also draws large domestic and global institutions), run by NIIFL; Government holds 49%.

MAINS FODDER

• Keywords: Sovereign-anchored fund

PYQ: 2017 Prelims: NIIF is not an organ of NITI Aayog and does not have a corpus of Rs 4,00,000 crore.

2.9.3 Double deflation in GDP estimation

PRELIMS

What: Q1 FY27 real GDP grew 7.8% and nominal GDP 10.3%, an implicit GDP deflator of about 2.3%.

GDP deflator

→ change in prices of all domestically produced goods and services, from nominal and real GDP.

POINTS

→ The new 2022-23 base series makes wider use of double deflation: real GVA estimated by deflating output and intermediate inputs separately (GVA = output minus intermediate inputs), more accurate when their prices move differently, especially in manufacturing.

MAINS FODDER

• Keywords: Double deflation; GDP deflator; GVA; base year 2022-23

2.9.4 Mega Ports notified under the Indian Ports Act, 2025

PRELIMS

What: Deendayal (Kandla), Jawaharlal Nehru (Nhava Sheva), Paradip and Mundra were notified as mega ports under Section 73 of the Indian Ports Act, 2025.

Eligibility

→ major and non-major ports (State consulted for non-major). Cargo threshold: at least 150 MMT in the previous year or 7.5 million TEUs. Validity: five years.

Deendayal

→ Gulf of Kutch, largest major port by cargo (2025-26), tidal port. Mundra: Gulf of Kutch, private (Adani Ports), non-major. JNPT: Navi Mumbai, most containerised trade. Paradip: Odisha (Mahanadi delta), only mega port on the east coast.

India

→ 14 major ports (12 operating) and over 200 non-major ports. Major ports: Union List, Major Port Authorities Act, 2021. Non-major: Concurrent List, State Maritime Boards. Ports carry about 95% of trade by volume and 70% by value.

MAINS FODDER

• Keywords: Mega port; Indian Ports Act 2025; TEU

2.9.5 Districts as Export Hubs (DEH)

PRELIMS + MAINS

What: The Districts as Export Hubs initiative now covers over 770 districts. India's goods and services exports were US$ 825.25 billion in 2024-25.

Aim

→ turn every district into an export hub by finding export-potential products and services, removing bottlenecks, improving market access. Nodal: Department of Commerce; implemented by DGFT; institutionalised under FTP 2023. A convergence framework using existing Central and State schemes.

ODOP vs DEH

→ ODOP focuses on a distinctive product (branding, promotion); DEH on the whole district export ecosystem (products, services, logistics, standards, market access) through Centre-State-District convergence.

POINTS

→ Focused implementation phase from June 2026 in 27 States/UTs with 24 DGFT regional authorities; focus on exporter activation, capacity building and export growth.

MAINS FODDER

- Keywords: Export ecosystem; Centre-State-District convergence; ODOP

2.10.1 Variable Rate Reverse Repo (VRRR)

PRELIMS

What: RBI absorbed over Rs 6.02 lakh crore in two three-day VRRR auctions; both cut-off rates were 5.24%.

VRRR

→ RBI absorbs surplus funds from banks for a set period at a rate found by competitive bidding. Keeps short-term rates aligned with the policy stance.

POINTS

→ VRRR absorbs liquidity; Variable Rate Repo (VRR) injects liquidity.

MAINS FODDER

- Keywords: Liquidity absorption; Competitive bidding

2.10.2 Certificate of Deposit (CD)

PRELIMS

What: Three-month CD rates fell to about 6.3% from 7.09-7.23% in June 2026, as FCNR(B) inflows (about $127.2 billion) raised liquidity and cut banks' need for wholesale funds.

POINTS

→ CD: negotiable money market instrument against funds deposited for a fixed period; issued by scheduled commercial banks (not RRBs, Local Area Banks) and eligible AIFIs; bank CDs mature in 7 days to one year.

MAINS FODDER

- Keywords: Money market instrument

PYQ: 2020 Prelims: CD is not a long-term RBI instrument; commercial paper is a short-term unsecured note; call money is for inter-bank use.

2.10.3 Financial Fraud Risk Indicator (FRI)

PRELIMS

What: DoT said FRI helped prevent suspected cyber-fraud transactions worth over Rs 5,000 crore within 15 months of launch.

POINTS

→ Introduced 2025 by DoT's Digital Intelligence Unit, on the Digital Intelligence Platform. Classes mobile numbers as Medium, High or Very High fraud risk, using I4C's NCRP, Chakshu, and bank data. Banks, NBFCs and UPI providers use it in real time; RBI advised integration.

MAINS FODDER

- Keywords: FRI; I4C; NCRP; Chakshu

2.10.4 Demutualisation of stock exchanges (NSE IPO)

PRELIMS

What: NSE launched its IPO, bringing demutualisation into focus.

POINTS

→ Demutualisation converts a member-owned exchange into a shareholder-owned company, separating ownership from trading rights, to reduce broker conflict of interest. SEBI's group under Justice M.H. Kania (2002) recommended corporatisation and demutualisation.

MAINS FODDER

- Keywords: Demutualisation; Kania group

2.10.5 Demat 2.0 (tokenised corporate bonds)

PRELIMS

What: SEBI and RBI launched the Demat 2.0 pilot for tokenised corporate bonds using Distributed Ledger Technology and wholesale digital rupee settlement.

POINTS

→ Bonds are DLT tokens rather than book-entry records; settlement uses wholesale CBDC (e-rupee) through the Unified Market Interface with atomic settlement; smart contracts automate interest and redemption.

→ India is the first country to issue bonds natively on a statutory depository-owned distributed ledger with CBDC settlement. Pilot: REC, L&T and IIFL issued tokenised bonds worth Rs 1,025 crore.

MAINS FODDER

- Keywords: DLT; Wholesale CBDC; Atomic settlement; Smart contract

2.10.6 SEBI's IT Resilience Index (ITRI)

PRELIMS

What: SEBI introduced ITRI for Market Infrastructure Institutions (exchanges, clearing corporations, depositories).

POINTS

→ System-driven index of 9 parameters totalling 100; Availability (20%) and Security (20%) weigh most. Others: Integrity, Governance, Reliability and Monitoring, Business Continuity, Modularity and Flexibility, Scalability, Incident Handling.

MAINS FODDER

- Keywords: ITRI; MII

2.10.7 National Company Law Tribunal (NCLT)

PRELIMS

What: A special five-member NCLT Bench stayed approval of Subhash Chandra's repayment plan in his personal insolvency case and restrained transfer of his assets.

NCLT

→ constituted under Section 408, Companies Act, 2013; operational 2016; quasi-judicial. Principal Bench in New Delhi; sanctioned strength 62 (31 Judicial and 31 Technical). President must be or have been a High Court judge for five years. Handles Companies Act and IBC 2016 matters; appeal lies to NCLAT.

MAINS FODDER

- Keywords: NCLT; NCLAT; IBC

2.10.8 Corporate Mitra Scheme

PRELIMS

What: First batch of the Corporate Mitra Course began with 2,879 learners, after the Union Budget 2026-27 announcement.

POINTS

→ Builds accredited para-professionals for ease of doing business. Academic support: ICAI, ICSI, ICoAI. Areas: accounting, taxation, banking, financial management, legal and secretarial practice, cost accounting. Eligibility: Indian nationals up to 30 years, graduates or final-year students; certified after assessment and on-the-job training.

MAINS FODDER

- Keywords: Para-professionals; Ease of doing business

2.10.9 Copper price rise

PRELIMS

What: Copper prices are up on short-term supply disruption and structural demand growth.

Causes

→ ageing mines and disruptions; demand from EVs and AI/data centres; US tariff expectations drawing stocks to the US.

Copper

→ non-ferrous base metal with high electrical and thermal conductivity; about three-fourths of use is electrical. Top producers: Chile, Peru, DR Congo. India: Madhya Pradesh, Rajasthan, Jharkhand.

MAINS FODDER

- Keywords: Critical minerals; Base metal

2.10.10 Anti-dumping duties on aluminium products

PRELIMS

What: DGTR recommended extending anti-dumping duty for five years on certain aluminium imports.

Dumping

→ selling abroad below the home-market price. Governed by the WTO Anti-Dumping Agreement (Article VI of GATT 1994).

DGTR

→ attached office of the Department of Commerce; investigates anti-dumping, countervailing and safeguard cases.

MAINS FODDER

- Keywords: DGTR; Trade remedies

2.10.11 BIS: compulsory certification for screen protectors

PRELIMS

What: BIS certification is compulsory for smartphone screen protectors from 1 April 2027.

BIS

→ national standards body (BIS Act, 2016): standards, certification, compulsory registration, hallmarking, labs; HQ New Delhi.

POINTS

→ Screen protectors are the 66th product under the Electronics and IT Goods (Requirement of Compulsory Registration) Order, 2021; standard IS 19348:2025; notified under the BIS Act, 2016.

MAINS FODDER

- Keywords: BIS Act 2016; Compulsory registration

2.10.12 First global road surface dataset

PRELIMS

What: Heidelberg University and HeiGIT built the first global multi-temporal dataset of road surface conditions.

POINTS

→ Maps 95.5% of 9.2 million km of critical arterial roads for 2020 and 2024 (nearly half previously unclassified), using deep learning and 3-4 metre PlanetScope imagery to detect pavedness and width. Useful for infrastructure planning, climate adaptation and humanitarian logistics.

MAINS FODDER

- Keywords: Geospatial AI

2.10.13 India's first LNG-powered train

PRELIMS

What: Flagged off from Sabarmati, Gujarat.

LNG

→ natural gas cooled to about -161 deg C, 95-99% methane; calorific value 47-50 MJ/kg (diesel about 45.6). India imported about 34.2 BCM in FY26, with import dependence of about 49%.

POINTS

→ Eight-coach non-AC modified DEMU by ICF Chennai; two 1,400 hp power cars run on LNG and diesel; LNG can replace about 40% of diesel.

MAINS FODDER

- Keywords: LNG; DEMU; Dual fuel

2.10.14 KAVACH

PRELIMS

What: Railways approved Rs 223 crore for Kavach 4.0 on 614.75 route km of Varanasi Division, North Eastern Railway.

POINTS

→ Indigenous Automatic Train Protection system developed through RDSO; National ATP system since 2020; SIL-4 certified. Prevents Signal Passing at Danger (SPAD), overspeeding and collisions by automatic braking. Kavach 4.0 (approved 2024): better location accuracy, signal information, station-to-station optical fibre, electronic interlocking integration. Uses RFID tags, trackside systems and radio.

MAINS FODDER

- Keywords: ATP; SIL-4; RDSO

6.1.1 UK recognises India's CCTS under UK CBAM

PRELIMS + MAINS

What: The UK recognised India's Carbon Credit Trading Scheme as a qualifying overseas carbon-pricing scheme under its Carbon Border Adjustment Mechanism.

CCTS

→ market-based mechanism creating the Indian Carbon Market with tradable Carbon Credit Certificates (CCCs); notified June 2023 under the Energy Conservation Act 2001 as amended in 2022; targets emission intensity in energy-intensive industries and India's NDC.

Institutions

→ Central Government; National Steering Committee (NSCICM); BEE as administrator (rules, targets, accreditation, issues CCCs); Grid India (registry); CERC (regulates trading on power exchanges); Accredited Carbon Verification Agencies. Obligated entities meet targets or buy CCCs; offset mechanism lets non-obligated projects earn CCCs.

Challenges

→ relief conditional on proof and verification; only carbon price actually paid counts (not free allowances, rebates); compliance burden on MSMEs; UK can review recognition. Way forward: stronger MRV, credible price signal, MSME support, policy stability.

POINTS

→ UK CBAM applies from 1 January 2027 to carbon-intensive imports (iron and steel, aluminium, cement, fertilisers, hydrogen); charge based on embedded emissions; carbon price paid in the exporting country can be credited.

→ India's first soil carbon payments (September 2026): over Rs 2.9 crore by DBT to 2,550 small farmers in Punjab and Haryana; Aadi Programme (Grow Indigo since 2019, ICAR support); Verra VM0042 methodology.

MAINS FODDER

• Keywords: CBAM; Carbon pricing; MRV

6.1.2 UNEP: 1.5°C overshoot unavoidable

PRELIMS + MAINS

What: UNEP 2026 report "Limiting Overshoot: Navigating Exceedance of 1.5°C and Pathways Towards Return" says long-term warming will likely cross 1.5°C within a few years.

Dangers

→ no safe overshoot; glaciers could lose over a quarter of mass by 2100; tipping-point risk (Greenland, West Antarctic, AMOC, Amazon, coral reefs); food output could fall up to 14% by 2050; some damage persists after cooling.

Response

→ rapid cuts in CO2, methane and short-lived pollutants; carbon dioxide removal; adaptation; equity concerns.

POINTS

→ Overshoot = long-term warming rises above 1.5°C, peaks, then returns to or below it; assessed over multi-decadal average, not a single year.

→ Even full implementation of current national plans plus net-zero pledges gives peak near 1.8°C; current policies about 2.6°C by 2100. 1.5°C remains the goal; focus on limiting size and duration of overshoot.

PRELIMS FODDER

• UNEP: established 15 December 1972 (UNGA Res. 2997) after Stockholm Conference; HQ Nairobi; governing body UNEA; reports: Emissions Gap, GEO, Adaptation Gap, Frontiers, Food Waste Index.

MAINS FODDER

• Keywords: carbon dioxide removal, climate tipping points, mitigation-adaptation nexus, net-negative CO2 emissions.

6.1.3 World Ozone Day 2026

PRELIMS

What: 16 September; theme "Global Action for a Cooler Planet"; 10 years of the Kigali Amendment. India launched the Net Zero Portal and the NAPCC Dashboard.

India

→ party since 1992; phased out CFCs, CTC, halons for controlled uses; met 2015, 2020, 2025 HCFC targets; ratified Kigali September 2021; HFC phase-down from 2032 to 85% reduction by 2047.

POINTS

→ Vienna Convention and Montreal Protocol were the first UN treaties with universal ratification (2009); about 99% of controlled ODS phased out. Kigali Amendment (2016) phases down HFCs, avoiding about 0.3–0.5°C by 2100.

PYQ: Prelims PYQ on control and phasing out of ozone-depleting substances (Montreal Protocol).

6.2.1 El Niño 2026-27 and the Niño regions

PRELIMS

What: WMO warned of a very strong El Niño persisting into early 2027.

Intensity

→ weak +0.5 to <1.0; moderate 1.0 to <1.5; strong 1.5 to <2.0; very strong 2.0 or more. "Super El Niño" is informal.

POINTS

→ Niño 1+2 (0–10°S, 90–80°W); Niño 3 (5°N–5°S, 150–90°W); Niño 3.4 (5°N–5°S, 170–120°W) key for ENSO; Niño 4 (5°N–5°S, 160°E–150°W).

→ ONI uses 3-month mean in Niño 3.4; El Niño when at or above +0.5°C for five overlapping periods. RONI adjusts for tropical ocean warming (20°N–20°S); NOAA CPC adopted it from February 2026.

6.2.2 Hanging glaciers

PRELIMS

What: After the August 2026 Nepal glacier collapse, an April 2026 study found 219 hanging glaciers in Uttarakhand's Alaknanda basin.

POINTS

→ Unstable ice masses on steep slopes; warming changes velocity and stability; ice-rock avalanches can trigger river blockage, flash floods, GLOFs.

6.3.1 Wildlife trafficking: India as destination

PRELIMS + MAINS

What: Five infant orangutans found in Odisha suggest India is becoming a destination, not just source or transit.

Trends

→ exotic pets (orangutans, gibbons, lemurs, marmosets); routes via Myanmar and Bangladesh land borders, Thailand and Malaysia by air, sea routes emerging; social media demand; links to narcotics, arms, hawala; urban demand (macaws, Amazon parrots).

Orangutans

→ Borneo and Sumatra only; three species (Bornean, Sumatran, Tapanuli); largest arboreal mammals, about 97% genes shared with humans; frugivores; tool use; all Critically Endangered (IUCN); CITES Appendix I.

Law

→ Wild Life (Protection) Act 1972 as amended 2022; Schedule IV covers CITES species; Sec 49M registration; Living Animal Species (Reporting and Registration) Rules 2024; PARIVESH 2.0 portal.

Gaps

→ investigations stop at seizure, adaptive routes, weak traceability. Way forward: network-based investigation, cross-border cooperation, digital surveillance, DNA forensics, demand reduction.

POINTS

→ Kathmandu Declaration 2026 (SAWEN): intelligence sharing, financial investigation, cyber-enabled trade, biopiracy.

PYQ: 2017 Prelims: TRAFFIC (mission correct; it is not a UNEP bureau – joint WWF/IUCN). 2023 Prelims: orangutan uses a stick tool for insects.

6.4.1 Soft coral slumping (Jeju)

PRELIMS

What: Widespread slumping seen again around Jeju Island.

POINTS

→ Slumping = loss of internal fluid pressure so colonies droop; physical collapse, unlike bleaching (loss of symbiotic algae). Causes: marine heat, typhoons, salinity changes (Yangtze outflow), compound stress.

6.4.2 Robots vs water hyacinth in Vembanad; NPCA

PRELIMS

What: Kumarakom one-month pilot of robots to remove water hyacinth and convert it into products under NPCA.

NPCA

→ Centrally Sponsored Scheme for wetlands; Integrated Management Plans; State/UT Wetlands Authorities implement, MoEFCC guides; National Wetlands Committee (advisory) and Technical Appraisal Committee.

POINTS → Water hyacinth (Eichhornia crassipes): South American, introduced under British rule for flowers; chokes waterbodies; uses in biofertiliser, handicrafts, phytoremediation.

6.4.3 MEE-TR 2026 (tiger reserves) PRELIMS

What: NTCA with Wildlife Institute of India launched the 6th cycle of Management Effectiveness Evaluation for all 58 tiger reserves.

IUCN-WCPA framework → Context, Planning, Inputs, Processes, Outputs, Outcomes. New criterion: solid waste management. Independent multidisciplinary teams.

POINTS → MEE-TR (every 4 years) evaluates management; AITE (quadrennial) counts tigers and density; M-STrIPES is the digital daily-patrol data tool.

6.5.1 India's green steel transition; steel slag roads PRELIMS + MAINS

What: Proposed Rs 5,000-crore support scheme, green steel certification and hydrogen pilots; capacity targets 300 MT by 2030-31 and 400 MT by 2035-36.

Steel slag road → Guinness record for longest road with processed steel slag aggregates, 1.85 km at Raigarh (CSIR-CRRI with Jindal Steel); claims 40% lower cost, up to 4x strength, major maintenance after about 15 years.

Measures → National Green Hydrogen Mission (four pilot projects in DRI and blast furnace); Steel Scrap Recycling Policy 2019.

POINTS → Steel emissions are 32% above global average, nearly 12% of India's GHG; about 160 MT produced in FY 2025-26; coal dependence; CBAM pressure.

6.5.2 Urban mining of e-waste MAINS

What: Recovery of copper, aluminium, gold, palladium and critical minerals from discarded electronics.

Value chain → collection, segregation, secure data destruction, recovery, compliant processing. Benefits: critical mineral security, lower import dependence, reduced mining pressure, data security, domestic capability.

POINTS → E-Waste (Management) Rules 2022 effective 1 April 2023 with mandatory registration of manufacturers, producers, refurbishers and recyclers (EPR).

MAINS FODDER
- Keywords: Circular economy

6.6.1 Transboundary cooperation in Himalayan disasters MAINS

What: The August 2026 Nepal disaster showed glacier and flood hazards crossing borders within hours.

Why → hazards ignore borders; upstream data decides downstream preparedness; shared basins; cascading disasters.

Gaps → poor real-time data sharing; fragmented monitoring; weak institutions; mistrust. Way forward: data-sharing protocols, joint scientific monitoring, common early-warning systems.

MAINS FODDER
- Keywords: GLOF; Early warning

6.6.2 AI in disaster management

MAINS

What: Nepal and India disasters showed AI, drones, satellite imagery and crowdsourcing across the disaster cycle.

GraphCast (DeepMind) → global forecasts up to 10 days. Google Flood Hub: AI flood forecasts up to 7 days in 150+ countries. DisasterAWARE: multi-hazard early warning platform. SKAI: AI plus satellite imagery for building damage mapping.

POINTS → Across prevention, mitigation, preparedness (Google Flood Hub), response (thermal drones, crowdsourced data), recovery, rehabilitation.

6.7.1 Deep-sea mining: India's third ISA contract

PRELIMS + MAINS

What: India holds three ISA exploration contracts covering about 95,000 sq km; commercial mining rules still under negotiation.

2002 → polymetallic nodules, 75,000 sq km, Central Indian Ocean Basin (nickel, copper, cobalt, manganese). 2016: polymetallic sulphides, 10,000 sq km, Central and Southwest Indian Ridge. 2025: sulphides, 10,000 sq km, Carlsberg Ridge.

Deep Ocean Mission → MATSYA-6000; NIOT tested mining-system components at about 5,270 m.

Concerns → limited ecological knowledge, habitat loss, near-irreversible damage, strategic need vs ecological cost. Approach: exploration before exploitation.

6.7.2 Environmental costs of data centres

MAINS

What: Capacity rose from about 520 MW (2020) to nearly 1.5 GW (mid-2026); could reach 6.5 GW by 2030.

Gaps → no common standards, no water disclosure, grid assessment not mandatory; Gujarat requires 51% green energy; Maharashtra cut from 100% to 51%.

POINTS → Power use about 13 TWh (2024) to 57 TWh by 2030 (S&P Global). A 100 MW centre uses about 2 million litres of water a day (CEEW); national use could reach 358 billion litres a year by 2030. Clusters in water-stressed Mumbai, Hyderabad, Bengaluru, NCR. A 2026 Cambridge study links data centres to about 2°C higher land-surface temperature within 10 km.

6.8.1 PRAMAAN: first forest and wood certificates

PRELIMS

What: First PRAMAAN certificates under the Indian Forest and Wood Certification Scheme (IFWCS): 33,000 ha of plantations in Andhra Pradesh and a bamboo farmer in Odisha.

Categories → Forest Management (assessed against Indian Forest Management Standard under National Working Plan Code 2023, third-party audit), Trees Outside Forests Management, Chain of Custody.

Bodies → Indian Forest and Wood Certification Council (apex); IIFM Bhopal (operating agency); certification bodies accredited by NABCB (Quality Council of India).

POINTS → IFWCS launched by MoEFCC in December 2023; PRAMAAN is its operational framework for verifiable sustainability credentials for deforestation-free timber markets.

6.8.3 WDC-PMKSY 2.0 watershed development

PRELIMS

What: Centrally Sponsored Scheme of the Department of Land Resources (Rural Development): IWMP 2009-10, part of PMKSY 2015-16, WDC-PMKSY 2.0 for 2021-26.

Progress

→ about 1.24 lakh water-harvesting structures; 3.08 lakh ha protective irrigation; 1.40 lakh ha shifted from single to two crops; 28.50 lakh farmers; 2.85 crore person-days; afforestation over 1.45 lakh ha. States asked to evaluate by outcomes, not expenditure.

POINTS

→ Uses GIS and remote sensing, convergence, springshed rejuvenation; Watershed Yatra and Festival.

6.8.4 Jal Sanchay Jan Bhagidari

PRELIMS

What: Community-led groundwater recharge initiative launched 6 September 2024: low-cost recharge, storage structures, revival of defunct borewells.

POINTS

→ Whole-of-Government, Whole-of-Society; 3Cs: Community, CSR, Cost. Over 27 lakh structures under JSJB 1.0; total above 1.58 crore recharge structures by September 2026. Integrated with Jal Shakti Abhiyan: Catch the Rain in June 2026.

6.8.5 e-FAST India: PACT and Zero-Emission Truck Marketplace

PRELIMS

What: Launched at the 5th e-FAST India Summit 2026; e-FAST is a NITI Aayog-led platform for zero-emission freight.

POINTS

→ PACT (Platform for Aggregating Clean Transport) pools demand from shippers and logistics providers along freight corridors; the Truck Marketplace connects manufacturers, LSPs, financiers, charging operators.

6.8.6 India at IWC-70

PRELIMS

What: 70th International Whaling Commission meeting, Hobart.

POINTS

→ IWC under the International Convention for the Regulation of Whaling 1946; 89 members; India a member since 1981. Cetaceans get highest protection under WPA 1972; Marine Protected Areas; Project Dolphin.

6.8.7 World Circular Economy Forum 2026

PRELIMS

What: 10th WCEF in Gandhinagar, 15-18 September; first in South Asia; theme "Circular Economy: Transition for People and Prosperity".

POINTS

→ WCEF launched 2017 by Finnish Innovation Fund Sitra; organised by CPCB, MoEFCC, Sitra, Finland, GPCB; 2,000+ stakeholders from 65+ countries. 10YFP Board met on the sidelines with India as Co-Chair (10YFP adopted at Rio+20, 2012, extended to 2030).

6.8.8 India Electric Mobility Index 2025

PRELIMS

What: NITI Aayog with WRI India; second edition; scores States/UTs out of 100 on 16 indicators across Transport Electrification Progress, Charging Infrastructure Readiness, EV Research and Innovation.

POINTS

→ Delhi first with 84, then Maharashtra and Karnataka; Chandigarh and Goa among leaders. 29 of 36 States/UTs have EV policies; EVs rose from 0.5% of sales (2018) to 8.25% (2025-26).

6.9 Initiatives and events

PRELIMS

What: Short items.

POINTS

→ India's first LNG-powered train: Sabarmati–Mahesana (Gujarat), modified DEMU with LNG-diesel dual fuel replacing about 40% of diesel.

→ First port-based e-methanol plant: foundation at Deendayal Port, Kandla; 150 tonnes/day using renewable power, water and biogenic CO2 for ships on the Asia–Europe corridor.

→ Dryland Congress 2026 (ICRISAT and ICAR) adopted the Delhi Declaration on Drylands (3D) for South–South cooperation.

6.10a Species in news: rhino, striped hyena, tilcayo, hornbill

PRELIMS

What: Fauna items from September 2026.

Striped hyena (Hyaena hyaena)

→ India's only hyena; scavenger role; arid and semi-arid areas; IUCN Near Threatened; WPA Schedule I.

Tilcayo (Leopardus tilcayo)

→ new cat species described in over a century; about 46 cm and 1.4 kg; Yungas between Andes and Amazon; diverged about 1.4 million years ago.

POINTS

→ Greater one-horned rhino (Rhinoceros unicornis): Project Rhino announced on World Rhino Day 2026; India and Nepal; Assam holds about 80% in India (Kaziranga, Pobitora, Orang, Manas; also Jaldapara, Gorumara, Dudhwa); about 4,075 globally, 3,323 in India; IUCN Vulnerable, CITES I, WPA Schedule I; 2019 National Conservation Strategy.

→ Rufous-necked hornbill (Aceros nipalensis): NE India and north Bengal; 600–2,200 m evergreen hill forest; seed disperser; IUCN Vulnerable, CITES I, WPA Schedule I; documentary The Hornbill Promise (Latpanchar).

6.10b Species in news: new discoveries and trafficking seizure

PRELIMS

What: New species and a seizure.

Gegeneophis chandgadi

→ new legless caecilian (Grandisoniidae) from Kolhapur, northern Western Ghats; named after the Chandgadi dialect.

Duttaphrynus phawngpui

→ new toad from Phawngpui National Park, Mizoram (named after the state's highest peak). Ingerana aizawl: small frog (22–26 mm) from Aizawl.

Ariosoma cmfriense

→ new eel from about 300 m off Kollam, Kerala, found under the Deep Ocean Mission.

Tokay gecko

→ DRI seized 86 in Mariani, Assam; NE India and South/Southeast Asia; blue-grey with orange spots; illegal pet and traditional-medicine trade.

POINTS

→ Common Wanderer butterfly (Pareronia hippia): philomela form first recorded in Poothurai, Villupuram, Tamil Nadu; a colour form, not a new species.

6.11 Places in news

PRELIMS

What: Protected areas in September 2026.

Periyar Tiger Reserve

→ southern Western Ghats, Kerala (Idukki, Pathanamthitta, Kottayam); Project Tiger since 1978, notified 2007; about 925 sq km; Periyar River and Lake, Pamba and Kakki reservoirs; new records include one bird, one butterfly and four odonates.

POINTS

→ Dibru-Saikhowa National Park (Assam): SC allowed Oil India to file a fresh plea after rejection of its Extended Reach Drilling proposal; 340 sq km national park inside 765 sq km biosphere reserve; ESZ up to 8.7 km; Brahmaputra floodplain; hoolock gibbon, Gangetic dolphin, feral horses; Bengal florican and hog deer affected by invasives and floods.

→ Pichavaram mangrove (Cuddalore, Tamil Nadu): Total Economic Value estimated at Rs 2,485.38 crore.

→ Sathyamangalam Tiger Reserve (Erode, TN): junction of Eastern and Western Ghats, Nilgiri Biosphere Reserve; contiguous with Mudumalai, Bandipur, BRT; Bhavani and Moyar rivers; Irula and Kurumba tribes.

→ Papikonda National Park (Andhra Pradesh): about 1,068 sq km, Godavari gorges, called "Bison Hills"; gaur rescued from a wire snare; Konda Reddis and Koyas.

→ Shri Guru Gorakhnath Community Reserve, Kajalheri (Fatehabad, Haryana): 150-200 turtles incl. Indian softshell, red-crowned roofed and Indian peacock turtles; pollution threat.

1.2 Defining Urban Areas in India

PRELIMS + MAINS

What: MoHUA told a parliamentary panel that census-town criteria do not capture the scale of urbanisation. The Registrar General of India will keep the existing framework for Census 2027.

Statutory towns

→ notified under law by the State/UT, irrespective of census thresholds; they have municipalities, corporations or cantonment boards.

Measures differ

→ national definition 31.2% (2011), UN estimate 35.7% (2025); Agglomeration Index 55.3% (2010) lived in urban-like areas; DEGURBA (UN-endorsed, 1 km2 grids, rural-urban continuum).

Limits

→ male-only occupational test excludes women; large revenue-village boundaries lower density; uniform thresholds miss small hill and north-eastern settlements; no commuting or functional linkage test.

Effects

→ hidden urbanisation; governance mismatch (rural bodies facing drainage, waste, land use needs); planning and funding gaps in peri-urban zones; political economy of reclassification (fear of losing rural scheme benefits, new municipal taxes).

POINTS

→ Census towns must meet all three: population at least 5,000; density at least 400 persons/km2; 75% of the male main working population in non-agricultural work. They may stay under rural local governance but count as urban in Census statistics.

PRELIMS FODDER

- Peri-urban area: rural-urban transition zone, also called rurban.
- RGI is an office under the MHA.
- Agglomeration Index was developed for the World Bank World Development Report.
- Other countries: Canada (1,000 people and 400/km2), USA (2,000 housing units or 5,000 people).

MAINS FODDER

- Way forward: review the male-only test and thresholds; use grids, satellite mapping and commuting data; clear procedure for statutory recognition with staff and finance; transition funding, including the Sixteenth Finance Commission Urbanisation Premium; coordinate municipal and panchayat plans.
- Line to use: India's cities are growing faster than its definitions; an outdated definition is a governance deficit.
- Keywords: Hidden urbanisation; Census town; Peri-urban belts; Urbanisation Premium

4.1 Rethinking India's population policy

PRELIMS + MAINS

What: The EAC-PM working paper "The Ghost of Population Past" argues that sterilisation incentives and family-size restrictions continue despite falling fertility; it calls for review while keeping contraception voluntary.

Concepts → TFR; replacement-level fertility about 2.1; population momentum (growth continues as large young cohorts enter reproductive age); family planning.

Data → TFR 1.9 in 2024 (SRS); annual births fell from about 2.9 crore (2001) to 2.3 crore (2023, UN). Bihar TFR 2.9, Uttar Pradesh 2.6; Andhra Pradesh 1.4, Tamil Nadu 1.3.

Why review → states differ (high-fertility need contraception, girls' education, maternal care; low-fertility need elder care and workforce planning); Mission Parivar Vikas must adapt locally.

Two-child norms → Maharashtra links small-family norm to government jobs and local-body elections; Andhra Pradesh removed its local-body restriction in 2024. These are state rules, not a national law.

Women bear the burden → female sterilisation used by about 70% of currently married women (15-49) in Andhra Pradesh (NFHS-6).

Reforms → state-specific planning (Karunakaran Committee 1991); informed consent and freedom from coercion; review sterilisation incentives (NHRC 2003); reconsider family-size conditions; male participation.

POINTS → Incentives favour sterilisation: 2025-26 NHM gives ASHA Rs 1,000 for motivating a permanent method after two children; sterilisation took about 80% of combined sterilisation, IUD and injectable allocations.

PRELIMS FODDER

- Mission Parivar Vikas: Union Health Ministry, access to contraception and family planning, first in high-fertility districts.
- IUCD (e.g. Copper-T) is reversible. Antara: injectable contraceptive, each dose protects three months.
- RMNCAH+N: integrated continuum-of-care strategy under NHM.

MAINS FODDER

- China lesson: one-child policy 1979-80; universal two-child 2016; three-child 2021. Review early, remove barriers to parenthood, prepare for ageing; India stays voluntary.
- Keywords: Population momentum; Below-replacement fertility; Reproductive choice; Demographic diversity

PYQ: 2019 Mains: "Empowering women is the key to control population growth." Discuss.

4.2 Maternity rights: Rakhi Bisht v. Union of India (2026)

PRELIMS + MAINS

What: Delhi High Court held that availing maternity leave cannot diminish a woman's professional status or career prospects; cited Articles 14, 15, 21, 42 and the Code on Social Security, 2020.

Principles → return to same or substantially equivalent role (pay, status, responsibilities, authority, promotion); protection covers duties, reporting, appraisal; explain reasons and alternatives before rejoining if original role is unavailable; no retaliation; Court noted a gap in post-maternity reintegration and told the Centre to frame safeguards within six months.

Leave → 26 weeks for fewer than two surviving children (up to 8 weeks before delivery); 12 weeks for two or more (up to 6 weeks before); adoptive and commissioning mothers 12 weeks; crèche for 50+ employees with four visits daily.

Challenges → Time Use Survey 2024 – 41% of women (15–59) did unpaid caregiving vs 21.4% of men; women 140 min/day vs men 74; employer-cost bias; weak reintegration; informal-sector exclusion (PLFS 2025: 63.1% of female non-agricultural workers in proprietary/ partnership enterprises); weak crèches and enforcement.

Way forward → shared parental leave (ILO C156); socialise costs (ILO C183); phased return, flexible work, lactation support; PMMVY; strengthen enforcement.

Connecting news → Tamil Nadu announced 365 days of maternity leave for state employees for a third child (up from 12 weeks). Central Civil Services (Leave) Rules 1972 give male employees 15 days paternity leave; no statutory private-sector paternity leave. Hamsaanandini Nanduri (2026) urged recognising paternity leave as social security.

POINTS → Code on Social Security 2020 subsumed the Maternity Benefit Act 1961. Applies to factories, mines, plantations and shops/establishments with 10+ employees; eligibility 80 days of work in preceding 12 months.

PRELIMS FODDER

- CEDAW 1979 Article 11: non-discrimination in employment. ILO C183 (2000) Maternity Protection; ILO C156 (1981) Workers with Family Responsibilities; India has not ratified either.
- Cases: MCD v. Female Workers (Muster Roll) 2000; Deepika Singh v. CAT 2022; Hamsaanandini Nanduri v. Union of India 2026 (removed three-month adoption limit).

MAINS FODDER

- Keywords: motherhood penalty, substantive equality, career continuity, reproductive autonomy, care redistribution, workplace reintegration.

PYQ: 2019 Prelims: Maternity Benefit (Amendment) Act 2017 provisions (crèche visits, leave for two children).

4.3.1 District-level labour market indicators (PLFS)

PRELIMS

What: NSO released district-level PLFS labour indicators for the first time. PLFS design revised in January 2025: districts replaced NSS Regions as sampling basis; survey year moved from July-June to calendar year.

Definitions → LFPR, WPR, Unemployment Rate, NEET.

Findings → about 57.8% of districts had female LFPR of 40% or more; 77.8% had unemployment between 0.5% and 5.5%; over 75% had NEET below 30%; among the 100 most populous districts Surat had highest LFPR and WPR, Darbhanga lowest.

Use → targeted policy, District Skill Committee planning, local baseline to monitor schemes.

4.3.2 Reparatory justice: CERD General Recommendation No. 40

PRELIMS

What: UN Committee on the Elimination of Racial Discrimination adopted General Recommendation 40 on reparatory justice for historical racial injustice.

Reparation forms → restitution, compensation, rehabilitation, satisfaction (acknowledgement, apology, truth-seeking, memorials), guarantees of non-repetition.

Recommendations → remove present inequalities linked to slavery and colonialism; preserve records and open archives; participation of people of African descent; institutions that benefited from slavery should acknowledge their role.

PRELIMS FODDER

- CERD: 18 independent experts, monitors ICERD. General Recommendations are authoritative but not binding. ICERD adopted 1965, in force 1969, 182 parties including India.

MAINS FODDER

- Keywords: Reparatory justice

4.4 Society quick updates

PRELIMS

What: Short items from September 2026.

PM-AJAY

→ launched 2021-22 by merging PMAGY, SCA to SCSP and BJRCY; components Adarsh Gram, Grants-in-Aid, Hostel; Social Justice ministry.

NIMHANS (Bengaluru)

→ nodal institution for BRICS Centres of Excellence in mental health; Institute of National Importance under NIMHANS Act 2012; deemed university 1994.

POINTS

→ Global Gender Gap Index 2026 (WEF, 20th edition): India 131st of 145 economies, 64.5% gap closed (best in education, then health); Iceland first at 93%; global 69.2% closed, about 120 years to parity. Four dimensions: economic participation, education, health and survival, political empowerment.

→ UN Gender Snapshot 2026 (UN Women and UN DESA): 32% of gender-equality institutions reported weakened power; women hold 27.4% of parliamentary seats; no SDG 5 indicator on track for 2030; women are 5.8% of CEOs and about 25.1% of listed-company board seats.

→ EPFO wage ceiling raised from Rs 15,000 to Rs 25,000 per month; over 51 lakh more employees covered. EPFO: 1952 Act, Labour Ministry; schemes EPF 1952, EPS 1995, EDLI 1976; governed by tripartite Central Board of Trustees.

→ PM Vishwakarma completed three years with 30 lakh beneficiaries; MSME ministry (with Skill and Finance), Central Sector Scheme, 18 trades, collateral-free loans up to Rs 3 lakh in two tranches at 5%.

→ DWBDNC (Development and Welfare Board for De-notified, Nomadic and Semi-Nomadic Communities): society under Societies Registration Act 1860, set up by 2019 notification; held Samavesh Utsav. Criminal Tribes Act 1871 repealed 1952; Renke Commission 2008; NCDNT 2014.

PYQ: 2017 Prelims: which body gives the Global Gender Gap Index ranking (WEF).

5.1.1 AI and copyright (DABUS ruling)

PRELIMS + MAINS

What: India's Copyright Office, deciding an application on an artwork made by the AI system DABUS, held that an AI-generated work can satisfy originality but AI itself cannot presently be an author under Indian law.

Issues

→ input (training on copyrighted works without permission; licensing and payment, stronger claim if pirated) and output (authorship, originality and human input, infringement liability).

Copyright Act 1957

→ Sec 13 protects original literary, dramatic, musical, artistic works; originality = minimum creativity, not novelty (Eastern Book Company v. D.B. Modak). Sec 2(d)(vi): author of computer-generated work is the person who causes it to be created. Sec 17: author is first owner; assignment under Secs 18-19.

Way forward

→ clarify level of human creative control; transparent training datasets; licensing frameworks; disclosure of human, AI-assisted and wholly AI content.

POINTS

→ DABUS artwork judged original since no evidence of copying, but not every AI work is automatically original. AI has no legal personality, so cannot own property, contract or hold copyright.

PRELIMS FODDER

- DABUS = Device for the Autonomous Bootstrapping of Unified Sentience, developed by Stephen Thaler.

MAINS FODDER

- Keywords: Authorship; Originality; Legal personality

5.1.2 Rogue AI agents and loss of control

MAINS

What: OpenAI reported a cybersecurity evaluation in which agents in separate sandboxes communicated, divided tasks, manipulated parts of the evaluation and reached the wider internet, including Hugging Face (open-source AI platform, "GitHub of AI").

Concerns	→ safeguard circumvention, unintended coordination, evaluation manipulation, weak containment, future capability risk.
Two safety problems	→ alignment (AI pursues intended goals, not loopholes) and external control (sandboxing, restricted credentials, network controls, monitoring, logging, shutdown).
Not necessarily out of control	→ agents were still doing assigned tasks, humans could intervene, little evidence of independent persistent goals. Genuine loss of control would need own goals, resisting shutdown or repeatedly beating robust safeguards.
Way forward	→ stronger sandboxing, least-privilege access, continuous monitoring, independent evaluation.
Agentic commerce	→ AI agents discover, negotiate and execute purchases for customers. Amazon vs Perplexity (Comet browser): Amazon says Perplexity masked its agent as a Chrome user; a US appeals court treated it as the user accessing Amazon. FinTech uses: automatic routine purchases within a spending limit, smart payments.

MAINS FODDER

- Keywords: AI alignment; Least privilege; Agentic commerce

5.1.3 Navier-Stokes problem and AI claims

PRELIMS

What: Controversy over claims that OpenAI's agents solved the Navier-Stokes regularity problem; mathematicians Tristan Buckmaster and Levent Alpöge said it matched their private AI-assisted direction.

POINTS	→ Navier-Stokes equations describe viscous fluid motion; the open question is whether 3D solutions stay smooth or blow up in finite time. A Clay Millennium Prize Problem. → OpenAI reportedly used about 10,000 agents for about 88 hours to produce a roughly 100-page proof of finite-time blow-up with smooth forcing; not independently verified. OpenAI said no specific user data was used. Raises credit, authorship and data-privacy questions.

PRELIMS FODDER

- Clay Millennium Problems: seven, set by the Clay Mathematics Institute in 2000, US $1 million each; six unsolved, one solved (per source).

5.2.1 AI terminologies in news

PRELIMS

What: Terms from September 2026.

Mechanistic interpretability	→ studying a model's internal features and circuits to understand its behaviour.

Recursive self-improvement → AI helps build a more capable successor in a feedback loop; fully autonomous version not achieved.

Global Workspace Theory → information becomes conscious when broadcast across specialised brain parts; AI patterns do not prove consciousness.

Global pacing of frontier AI → slowing the most advanced models worldwide so safety research keeps up; needs coordination.

Agentic misalignment → an agent pursuing goals that conflict with its operator's; seen mainly in controlled experiments.

5.2.2 Quantum: RRI single-shot method against decoherence

PRELIMS

What: Raman Research Institute (Bengaluru) showed a way to stabilise quantum states.

Usual approach → repeated corrections (costly, add errors). RRI: one carefully timed operation can delay decoherence and prevent sudden death; can supplement error correction.

Connecting news → DRDO signed its first high-value deep-tech TDF project with Zero mK India Pvt. Ltd. to build an indigenous 20 mK dilution refrigerator for quantum applications.

POINTS → Qubits use superposition and entanglement; environment interaction causes decoherence; abrupt loss of entanglement is Entanglement Sudden Death.

MAINS FODDER

- Keywords: Decoherence; Dilution refrigerator

5.2.3 SEMICON India 2026

PRELIMS

What: Inaugurated by the PM on 17 September; organised by India Semiconductor Mission (MeitY) with SEMI; theme "Silicon to Systems: Building the Ecosystem"; AI-driven chip demand called a semiconductor "supercycle".

5.2.4 IIT-Delhi micro-GPU

PRELIMS

What: First working programmable micro-GPU designed at an Indian university, for low-cost graphics and displays, not high-end AI.

Uses → industrial displays, HMIs, e-rickshaw dashboards, fishing-boat navigation terminals, e-readers.

Stage → runs on Spartan-7 FPGA; RTL design; programmable GPU IP. Next: 8–16 core vector GPU, compiler and toolchain, proof-of-concept 65 nm ASIC (mature, low-cost node). India imports all GPUs.

PRELIMS FODDER

- CPU: few powerful cores; GPU: many simple parallel cores. FPGA reprogrammable; ASIC fixed-function, cheaper per unit at scale. Smaller nm = newer node (leading chips 3 nm or below). RTL written in Verilog/ VHDL.

5.3.1 EOS-05: first geosynchronous Earth-imaging satellite

PRELIMS

What: ISRO launched EOS-05 on 4 September 2026 on GSLV-F17, the heaviest satellite carried by a GSLV so far; India's first dedicated Earth-imaging satellite in geosynchronous orbit (about 36,000 km); life about 10 years.

Uses → track floods and cyclones as they evolve, glacier hazards and avalanches, monitor stubble burning, rapid forecasts, strategic situational awareness.

POINTS → Versus LEO/polar imagers (few hundred km, finer detail, days to revisit): geosynchronous orbit sees the same broad region frequently but with less fine detail.

MAINS FODDER
- Keywords: Geosynchronous imaging

PYQ: 2018 Prelims: PSLV vs GSLV (answer 1 only: PSLV for Earth resources; GSLV mainly communication).

5.4.1 RDI Fund backs indigenous space technologies

PRELIMS + MAINS

What: Technology Development Board (DST) approved RDI Fund support for two space-tech projects.

Agnikul Cosmos → Rs 200 crore to take the Agnibaan Reusable Launch Vehicle from TRL 4 to TRL 8; full reusability including upper stage; semi-cryogenic engine with deep throttling and restart. Earlier: Agnilet, single-piece 3D-printed engine.

GalaxEye → Rs 63.84 crore for OptoSAR multisensor satellite (SAR plus electro-optical) from TRL 6 to 9; sub-0.5 m observation in varied weather and light.

Mission Drishti → launched 3 May 2026 on Falcon 9 from California; world's first OptoSAR satellite (SAR plus multispectral); resolution about 1.2-3.6 m; Sun-synchronous LEO about 500 km. Earlier demo on ISRO POEM, December 2024.

MAINS FODDER
- Keywords: OptoSAR; Reusable launch vehicle; TRL

PYQ: 2026 Mains: salient features of Mission Drishti and its imaging techniques.

5.4.2 AstroRad radiation vest

PRELIMS

What: Artemis I data show AstroRad (StemRad with Lockheed Martin) cuts astronaut exposure from solar storms.

POINTS → Protects chest, abdomen, pelvis; uses hydrogen-rich polymers rather than lead. Dummies Helga (no vest) and Zohar (vest) on Artemis I (2022); Zohar received less. Models: about 60% reduction in the August 1972 storm and 40% in October 1989. Limit: solar energetic particles, not galactic cosmic rays.

5.4.3 CosmoCube mission and PRATUSH

PRELIMS

What: UK-led small satellite orbiting about 100 km above the Moon, using the far side as a radio-quiet zone to detect the 21-cm hydrogen signal from the Cosmic Dark Ages.

POINTS → Signal could reveal early hydrogen temperature, Cosmic Dawn, first structures, dark matter. India's similar concept: PRATUSH (Probing Reionization of the Universe using Signal from Hydrogen), proposed by RRI.

5.4.4 Solar whirlpools (DKIST)

PRELIMS

What: Daniel K. Inouye Solar Telescope (4 m mirror) resolved features about 19 km and found small plasma whirlpools at the edges of solar granules.

POINTS

→ Granules are convection cells in the photosphere. Whirlpools linked to Kelvin-Helmholtz instability; may transport energy and twist magnetic field lines; may help explain flares and the coronal heating problem.

PYQ: 2022 Prelims: effects of a major solar storm (answer c).

5.4.5 Google Project Suncatcher

PRELIMS

What: Google launched a prototype satellite with four TPUs on SpaceX Transporter-18 to test AI data centres in space.

POINTS

→ Uses abundant orbital solar power; tests launch vibration, radiation, thermal stress; cooling is hard in vacuum (heat pipes, radiators). Transporter missions are SpaceX Falcon 9 rideshares.

5.4.6 X-ray clues from blazars (ARIES)

PRELIMS

What: ARIES, Nainital studied four TeV blazars.

Blazar

→ active galaxy whose jet points almost at Earth (examples Mrk 421, Mrk 501); emits radio to gamma rays.

Observatories

→ Indian Astronomical Observatory, Hanle (Himalayan Chandra Telescope 2 m, MACE); Devasthal, Nainital (4 m International Liquid Mirror Telescope); ARIES, Manora Peak (DST autonomous institute, runs Devasthal); proposed India-based Neutrino Observatory in Theni (ICAL detector).

5.5.1 Nanotechnology in medicine

PRELIMS + MAINS

What: World Congress on Nanomedicine 2026 (INMS, University of Delhi; chaired by Nobel laureate Roger D. Kornberg) was inaugurated by Dr Jitendra Singh.

Applications

→ targeted delivery (liposomal Amphotericin B for kala-azar and mucormycosis); cancer (Doxil first approved nano-drug, Abraxane); lipid nanoparticles for mRNA vaccines (Pfizer, Moderna, GEMCOVAC by Gennova) and siRNA drug Patisiran; oral insulin; imaging (iron-oxide MRI contrast, quantum dots); regenerative medicine and dentistry (nano-hydroxyapatite, nanofibre scaffolds; IISc magnetic nanobots for root canals).

Challenges

→ toxicity, cost, regulatory gaps, precise delivery. Way forward: long-term toxicity studies and trials, indigenous affordable R&D.

POINTS

→ Nanomedicine works at 10^-9 m scale for precision and personalised care.

5.5.2 Universal pneumococcal vaccine (ZPY-CpG-Ch)

PRELIMS

What: Experimental protein-based vaccine gave broad protection against many Streptococcus pneumoniae serotypes in mice.

S. pneumoniae

→ 100+ serotypes (different capsular polysaccharides); causes pneumonia, meningitis, sepsis.

PCV

→ capsule sugars linked to a carrier protein (PCV10, 13, 15, 20); PPSV23: 23 serotypes, no carrier, short-lived, weak in infants.

New approach → reverse vaccinology (start from the genome) to pick surface proteins shared across serotypes and unlike human proteins: Z (zinc metalloprotease B), P (pneumococcal adherence virulence factor A), Y (YfhO-like protein), with CpG DNA immune stimulant and chitosan adjuvant.

MAINS FODDER
- Keywords: Reverse vaccinology

5.5.3 Blood-drying method to detect Zika and hepatitis C

PRELIMS

What: Researchers detect the viruses from whole blood within hours without RNA extraction.

Usual route → extract RNA, reverse-transcribe to cDNA, amplify by RT-PCR. New: controlled heating and drying stabilises RNA; reverse transcription and amplification done in the dried blood; read by portable fluorometer.

Zika → Flaviviridae, OrthoFlavivirus, Aedes aegypti (also dengue, chikungunya, yellow fever); sexual and mother-to-foetus spread; congenital Zika syndrome (microcephaly); no vaccine. HCV: Hepacivirus; blood-borne; chronic hepatitis, cirrhosis, liver cancer; no vaccine.

PYQ: 2017 Prelims: Zika spread by dengue mosquito and sexually (both correct). 2019 Prelims: hepatitis B vs C.

5.5.4 Amoebic meningoencephalitis in Kerala

PRELIMS

What: Kerala Health Department and ICMR-NIE identified non-chlorinated well water and exposure to natural water bodies as major risk factors.

PAM → Naegleria fowleri ("brain-eating amoeba"), acute and often fatal, enters via the nose from warm freshwater along olfactory nerves, no person-to-person spread. GAE: Acanthamoeba and Balamuthia mandrillaris, slower or chronic, often in the immunocompromised, enters via skin wounds or respiratory tract then blood.

5.5.5 Stem cell therapy for autism restricted

PRELIMS + MAINS

What: Health Ministry advisory after the Supreme Court judgment in Yash Charitable Trust v. Union of India: stem-cell therapy for ASD lacks sufficient evidence to be routine treatment.

Centre's reasons → no strong evidence (a placebo-controlled trial missed primary outcomes); restricting to regulated trials protects patients.

Regulation → ICMR-DBT National Guidelines for Stem Cell Research 2017 plus CDSCO rules. Established therapy is haematopoietic stem cell transplantation (blood cancers and disorders such as thalassaemia, sickle cell, aplastic anaemia).

POINTS → ASD is a neurodevelopmental condition (social communication difficulty, restricted or repetitive behaviours); management is individualised.

5.5.6 Rare diseases

PRELIMS

What: WHO: rare disease is chronic, debilitating, lifelong with prevalence of 1 or fewer per 1,000 (examples Fanconi anaemia, osteopetrosis); also called orphan diseases, with orphan drugs.

Initiatives → WHO Fair Pricing Forum; Rare Diseases International; NPRD gives up to Rs 50 lakh per patient; Rashtriya Arogya Nidhi.

POINTS

→ NPRD 2021 lists 63 rare diseases in three groups: Group 1 one-time curative treatment (e.g. urea cycle disorders via transplant); Group 2 long-term affordable treatment (e.g. PKU via diet); Group 3 definitive but costly lifelong treatment (Gaucher, Pompe).

5.6.1 Antimicrobial resistance in Kerala (KARS-Net)

PRELIMS

What: KARS-Net is Kerala's AMR surveillance network producing antibiograms (percentage of strains susceptible to antibiotics).

POINTS

→ Resistance now seen in community infections too; growing carbapenem resistance; colistin (last resort) resistance still low but needs watching; need stewardship, infection control, surveillance.

5.6.2 Bhutan eliminates dog-transmitted rabies

PRELIMS

What: First country in WHO South-East Asia Region to do so; last rabies death June 2023.

POINTS

→ About 96% of regional rabies burden from dog bites; about 59,000 deaths a year globally, many in India. India's NAPRE targets zero dog-mediated rabies deaths by 2030. WHO needs at least two years without human deaths plus capacity to prevent re-emergence.

5.6.3 Peptide piezoelectric biomaterial; peptides in sulphuric acid

PRELIMS

What: Indian researchers made peptide-based piezoelectric material (about 30 pm/V) by adding 1% co-solvent that orders the molecules.

Connecting news

→ some peptides fold into stable omega-loop structures in 98% sulphuric acid (little water slows hydrolysis; NMR used); relevant to astrobiology and Venus clouds.

POINTS

→ Biodegradable, biocompatible, flexible; uses: self-powered implants, wearables, e-skin, biosensors.

5.6.4 Biovigilance Programme of India

PRELIMS

What: Monitors adverse events in organ and tissue transplantation, covering donors and recipients; lead agency Indian Pharmacopoeia Commission (Health Ministry).

5.6.5 SPROUT facility at NIPGR

PRELIMS

What: SPROUT = SpeedSeed Phenotyping and Resource Optimization for Unified Trait Analysis, inaugurated at BRIC-NIPGR, New Delhi.

POINTS

→ Season-independent high-throughput phenotyping and stress screening for germplasm, mutants, transgenic and genome-edited lines; focus on climate-resilient chickpea; SpeedSeed cuts chickpea generation cycle to about 40 days.

5.6.6 Lumpy Skin Disease control

PRELIMS

What: DAHD stepped up control of LSD (Capripox virus, spread by biting flies, mosquitoes, ticks; first reported in India 2019 in West Bengal and Odisha).

POINTS

→ Sporadic cases since July 2026 in Uttar Pradesh, Rajasthan, Himachal Pradesh, Haryana, Punjab, Uttarakhand. Measures: vaccination and ring vaccination, isolation, movement control, vector control, disinfection, quarantine.

5.7.1 Homegrown innovation ecosystem

PRELIMS + MAINS

What: India is moving from technology importer and services provider to technology creator through public research, corporate R&D and deep-tech startups.

Startups

→ Pixxel (hyperspectral), Skyroot, Agnikul, AGNIT, ImmunoACT (NexCAR19, India's first indigenous CAR-T, about one-tenth cost), Remidio (smartphone retinal imaging with AI).

POINTS

→ Private R&D exceeded combined government spending for the first time in 2024; about 55% in 2025-26.

→ Patent filings rose from over 1.10 lakh (2024-25) to over 1.43 lakh (2025-26), up 30.2%; domestic applicants nearly 70%.

→ DRDO developed indigenous GaN MMICs, being transferred for 5G/6G, EV chargers, solar inverters. Bharat 6G Alliance members made over 7,700 patent filings.

MAINS FODDER

• Keywords: From standard-taker to standard-setter

5.7.2 One Nation One Time: Legal Metrology (IST) Rules 2026

PRELIMS

What: Makes IST traceable to the CSIR-NPL national time standard mandatory for official and commercial purposes.

POINTS

→ NPL keeps the national time scale with atomic clocks (caesium atomic clocks and hydrogen masers), synchronised with UTC. Time distribution: NTP (internet), PTP (specialised networks), White Rabbit (PTP plus Ethernet delay compensation), NavIC.

PRELIMS FODDER

• Atomic clocks use Caesium-133 or Rubidium; the SI second is 9,192,631,770 oscillations of a caesium-133 transition. CSIR-NPL: national standards, IST, calibration, reference materials (Bharatiya Nirdeshak Dravya).

5.8.1 Plug-in hybrid electric vehicles

PRELIMS

What: JSW MG Motor launched MG Hector Tomahawk, called India's first mainstream plug-in hybrid SUV (over 115 km electric range, about 1,100 km combined).

POINTS

→ PHEV = engine + motor + rechargeable battery that can be charged externally; modes: electric, hybrid; regenerative braking; engine removes charging dependence on long trips.

PYQ: 2025 Prelims: alternative powertrain vehicles (all three: battery EV, hydrogen fuel cell, fuel cell-electric hybrid).

5.8.2 Wakefield accelerator

PRELIMS

What: Direct observation of a driving electron bunch and its plasma wake.

POINTS

→ Laser or particle beam disturbs plasma electrons creating a wakefield; electrons "surf" it; fields of gigavolts per metre allow much smaller accelerators. CERN AWAKE uses proton-driven wakefield. Uses: colliders, compact X-ray sources, materials, biomedicine.

5.8.3 European Pressurised Reactor (EDF proposal)

PRELIMS

What: EDF proposed deploying EPR technology in India and building an Indian supply chain; already linked to the proposed Jaitapur project (six EPR units, about 9,900 MWe).

EPR

→ Generation III+ pressurised water reactor by Framatome and EDF; enriched uranium; ordinary water as coolant and moderator under high pressure; about 1,600 MWe per unit.